Advanced Programming 2025

Predicting International Football Success from Pre-Tournament Data

Final Project Report

Dany Gonçalves Rodrigues
dany.goncalvesrodrigues@unil.ch
Student ID: 21423926

January 9, 2026

Abstract

This project investigates the use of machine learning techniques to analyze and predict outcomes in major international football tournaments using pre-tournament performance data. A comprehensive feature set is constructed by combining diffenrent types of indicators based on Elo ratings, recent form metrics derived from historical match results, and group-level contextual variables reflecting competitive difficulty. Multiple predictive models are trained and evaluated using a Leave-One-Tournament-Out validation framework to ensure a realistic forecasting setup that relies only on information available before each tournament. Particular attention is given to probabilistic modeling and tournament-level consistency, allowing team outcomes to be compared in a structured and interpretable way. The project provides a probability-based framework for forecasting international football tournaments and applies the best trained model to generate predictions for the 2026 FIFA World Cup.

Contents

1 Introduction

Football is the most popular sport in the world, captivating billions of people across continents. Every few years, international tournaments such as the FIFA World Cup, UEFA Euro, Copa América, and the Africa Cup of Nations bring together nations and fans in a global celebration of competition and pride. For a few weeks, matches are watched, discussed, and relived across stadiums, television screens, and social platforms worldwide. Alongside the matches themselves, predicting which teams will succeed, surprise, or disappoint becomes a key part of the tournament experience, fueling discussions, expectations, and storylines among fans and analysts.

1.1 Problem Statement

Over the past decade, football analytics has become an increasingly important tool for performance evaluation and strategic decision-making. In international tournaments, expectations are shaped by past success, recent form, expert analysis, and data-driven evaluations. While these sources provide valuable signals, they also reflect differing assumptions and rarely offer a unified or transparent framework for assessing tournament success.

Despite these advances, predicting outcomes in international tournaments remains particularly challenging. Unlike club football, national teams compete infrequently, and their preparation for major tournaments is often based on a limited number of matches spread over long periods of time. This lack of consistent and comparable data makes reliable prediction more difficult, and commonly used ideas such as “form” or “momentum” are often discussed without a clear or quantitative definition.

This project aims to address these challenges by developing an objective and reproducible framework to quantify teams’ pre-tournament performance and examine how it relates to tournament outcomes. While discussions about predicting success in international tournaments are often driven by fan passion, intuition, and national loyalty, this work explores whether data can offer a more reliable alternative. Can pre-tournament information be used to meaningfully predict tournament outcomes, and which modeling approach is best suited to this task?

1.2 Objectives

The main objective is to develop and evaluate predictive models capable of estimating a national team’s success in international tournaments using only data available before the competition begins. To this end, the project constructs a comprehensive dataset that includes pre-tournament performance indicators such as win rates, goal differences, and opponent quality, as well as contextual variables describing the groups composition and relative team strength.

Using this dataset, three modeling approaches are evaluated: multinomial logistic regression, random forest, and XGBoost. In addition, a random guessing strategy is used as a baseline to provide a reference level of performance. Logistic regression is included as a simple classification model, while the two tree-based methods introduce increasing levels of model complexity and non-linearity. All approaches are assessed using a Leave-One-Tournament-Out (LOTO) validation strategy, enabling a consistent comparison of predictive performance across multiple international tournaments.

In addition to comparing modeling approaches, the project tries to identify which features are most influential in predicting tournament success. The best resulting framework is then applied to generate predictions for the 2026 FIFA World Cup.

2 Methodology

2.1 Data description

The first step of the project is to collect a comprehensive dataset of international football matches played by national teams. The data comes from the International Football Results database, which includes matches from 1872 to 2025 and is publicly available on platforms such as Kaggle and GitHub. Two datasets are used in this study: results and shootouts. The results dataset is the main source of information and contains

48,850 matches described by nine columns, including the match date, home and away teams, scores, and tournament name. Although the dataset is regularly updated as new matches are played, this does not affect the approach or conclusions of the project thanks to the methodology used. The shootouts dataset contains 657 matches and provides additional information on games decided by penalty shootouts, including the teams involved and the winner.

Once the data is downloaded, both datasets are cleaned to ensure consistency and usability. All date fields are converted to a standard datetime format, duplicate rows and fully empty records are removed, and matches with missing team names are discarded. In the shootouts dataset, many entries contain missing values for the first_shooter column; however, since this information is not needed for the analysis, the column is removed and does not affect data quality. Other unnecessary columns in the results dataset are also dropped. Finally, to focus on the modern era of football, all matches played before the year 2000 are removed. This process produces two versions of each dataset: a fully cleaned dataset covering all years and a filtered dataset restricted to matches from 2000 onward. These processed datasets serve as the foundation for all later feature engineering, modeling, and prediction steps in the project.

2.2 Dataset creation

The original datasets contain match-level results but do not directly describe how teams performed in entire tournaments. To predict tournament outcomes, additional processing is therefore required to construct tournament-level datasets based on the available match results.

The tournament_builder.py script generates a structured summary of major international football tournaments using the cleaned match results dataset. It filters the data to retain only key competitions including the FIFA World Cup, UEFA Euro, Copa América, African Cup of Nations, AFC Asian Cup, and Gold Cup after standardizing tournament names to lowercase to ensure consistency. For each tournament edition, the script groups matches by tournament and year in order to extract the corresponding start and end dates. Additional information about the tournament, including its confederation and continent, is also added. The resulting file, tournaments_info.csv, provides a concise and well-organized overview of all tournament editions and serves as a foundational reference for subsequent data processing and modeling stages.

The tournament_results_builder.py script generates tournaments_results.csv, a tournament-level dataset that summarizes each team's performance and final outcome in major international competitions. For each tournament edition, the script gathers all matches within the tournament's start and end dates and computes per-team statistics such as number of matches played. It then assigns each team a tournament stage (e.g., Champion, Runner-up, Semi-Finalist) based on a combination of match progression and final-round identification. This stage classification relies on the assumption of standard knockout tournament structures, inferring progression from match chronology and the total number of matches played per team as it will be explained in the next paragraph. Moreover, editions with irregular tournaments formats are excluded to ensure consistent and reliable stage assignment.

Since a final is the last game of a tourney, it is found by its the date within the tournament. Its winner is determined either from the match score or, if the final ends in a draw, using penalty shootout data. The losing finalist is labeled Runner-up. To classify the remaining teams, the script uses the maximum number of matches played by a team in that tournament as a reference: teams reaching one match fewer than the maximum are considered Semi-Finalists (max - 1), followed by Quarter-Finalists (max $-$ 2), Round of 16 (max $-$ 3), and Group Stage teams for earlier eliminations. Finally, results are sorted by tournament, year, and stage for readability. The resulting tournaments_results.csv dataset provides a crucial foundation for the modeling stage, because now we have the real outcome of each tournament.

The group_stage_composition_builder script generates group_stage_composition.csv by inferring group-stage assignments from match results and tournament information. For each tournament, it identifies each team's first three opponents as a proxy for group-stage matches. Teams that share the same opponents are assumed to belong to the same group and are assigned a group label (A, B, C, etc.).

2.3 Features

We now turn to the three main categories of features used in the predictive modeling phase: Elo ratings, pre-tournament performance metrics, and group-related metrics. First, the Elo rating is used to capture the overall competitive strength of a team entering a tournament. The elo_calculator.py script computes dynamic Elo scores for each national team using historical international match results from 2000 onward, updating each teams' elo after every match played.

Let R_Aand R_B denote the pre-match Elo ratings of teams A and B. All teams start with an elo rating of 1500 at 2000-01-01. To account for home advantage, the home team's rating is first adjusted by a constant offset H = 100 :

$$R_A^* = R_A + H$$

Expected win probabilities are then computed using this adjusted rating:

$$E_A = \frac{1}{1 + 10^{\frac{R_B - R_A^*}{400}}}, \quad E_B = 1 - E_A$$

Match outcomes are computed numerically as S=1 for a win, S=0.5 for a draw, and S=0 for a loss. New elo ratings are then updated according to:

$$R'_A = R_A + K(S_A - E_A), \quad R'_B = R_B + K(S_B - E_B)$$

where K = 30 is a fixed sensitivity parameter controlling the magnitude of rating adjustments.

Finally, Elo ratings computed for individual matches are summarized at the team level, allowing each team's rating to be identified just before the tournament start date. This pre-tournament Elo rating is used as a key measure of team strength in the prediction models.

The second category of features, pre-tournament performance metrics, captures each team's recent form leading up to a major competition. Using the features.py script, these indicators are computed from the twelve most recent matches played by each national team before the start of the tournament. For every team, the code filters all games occurring before the competition's start date and computes several quantitative measures that summarize their recent performance on the last 12 games. Those are the win rate, the goals scored/match, goals conceded/ match, the goal difference and finally, the average opponent Elo which provides an estimate of the quality of opposition faced during this period, ensuring that form is weighted by the relative strength of competitors. Together, these indicators describe both the efficiency and resilience of each team before a tournament.

The third category of features, group-level metrics, captures contextual information about each tournament's group composition. By combining pre-tournament data with group-stage assignments, the model computes indicators such as the average and maximum Elo within each group, the difference between a team's Elo and the group average, and each team's Elo rank within its group.These metrics collectively measure the relative difficulty of each group, allowing the model to capture contextual factors that influence advancement probability, such as facing multiple top-tier opponents in the same pool.

2.4 Models

This section presents the modeling approach used to predict tournament outcomes based on the feature set developed earlier in the project. These features summarize each team's pre-tournament strength, recent performance, and competitive context, and are used to estimate how far national teams will progress in a given competition.

The prediction task is formulated as a multi-class classification problem in which the target variable corresponds to the tournament stage **reached by each team.** Outcomes are encoded into six ordered categories: Champion, Runner-up, Semi-Finalist, Quarter-Finalist, Round of 16, and Group Stage. Model performance is also evaluated using a reduced three-category grouping (**Deep Run** = Champion, Runner-up and Semi-finalists, **Knockouts** = Quarter-finalists and round of 16 and **Group Stage**) (see Figure 5a, 5b in the Appendix)referenced as macro3 which provides a more stable and interpretable assessment of predictive accuracy.

Model training and evaluation follow a Leave-One-Tournament-Out (LOTO) strategy that closely mirrors a real-world prediction setting. Each tournament from 2016 onward is used as a test case, while all earlier tournaments from 2000 to 2015 serve as training data. This process ensures that predictions are always based solely on historical information and **avoids information leakage across competitions.**

Within this evaluation framework, model predictions take the form of probability distributions over all possible tournament stages for each team **individually**. To convert these team-level probabilities into coherent tournament-level predictions, a **tournament success score** is computed. This score is defined as a weighted sum of the predicted probabilities, with higher weights assigned to deeper stages of the tournament and lower weights to early eliminations as follows:

[Tournament Success Score = 6 ×P(Champion) + 5 ×P(Runner-up) + 4 ×P(Semi-finalist) + 3 ×P(Quarter-finalist) + 2 ×P(Round of 16) + 1 ×P(Group Stage)]

Predicted stages are then assigned according to this ranking while respecting the tournament structure, such as enforcing a single champion and runner-up, ensuring that predictions remain coherent and comparable.

Using this framework, three classification models of increasing flexibility are trained and evaluated: multinomial logistic regression, random forest, and gradient boosting (XGBoost). All three models are trained on the same feature set and are designed to estimate, for each team, the probability of reaching each tournament stage. The modeling approaches include multinomial logistic regression as a **linear model**, alongside random forest and XGBoost, which capture non-linear relationships **using tree-based ensembles.**

2.5 Models Evaluation

Model performance is then evaluated using three complementary metrics. **Stage accuracy** measures if the prediction matches the outcome computed binary as : If Prediction = Reality = 1 and if Prediction $\neq$ Reality = 0. Then we compute its mean to get the percentage of right stages reached by teams predicted. **Macro3 accuracy** is calculated the same way. **Mean Absolute Error (MAE)** quantifies the average distance between predicted and true tournament stages. An average MAE of 1 indicates that predictions are, on average, off by one stage (e.g., semi-final instead of quarter-final), which means the lower the average is, the better. From those we get those results about our 3 models:

Model	Stage Accuracy	Macro-3 Accuracy	MAE (Stage)
Logistic Regression	**0.47**	**0.67**	**0.80**
Random Forest	0.42	0.66	0.88
XGBoost	0.39	0.62	0.97

Table 1: Predictive performance of evaluated classification models (LOTO average)

The comparative evaluation highlights some differences in performance across the three models. Multinomial logistic regression achieves the highest average stage accuracy (0.47) and macro-3 accuracy (0.67), while having the lowest mean absolute error (0.80), indicating more precise and stable predictions overall. The random forest model performs slightly worse across all metrics and the XGBoost shows the weakest performance. This could be explained by the relatively limited number of historical tournaments and the low number of features, which reduce the benefits of more complex, non-linear models and increase their susceptibility to overfitting. As a result, the Logisitc Regression will be used to predict the World Cup 2026 as it is our best predictive model.

3 Implementation

The project is structured as a modular Python pipeline within the src/ directory, with separate modules for data downloading, data preparation, feature engineering, model training, and visualization (see Figure 7 Project structure).

3.1 Key technical decisions

Different implementation challenges appeared due to the structure of international football data. A key concern was preventing information leakage, as all features needed to be computed using only data available before each tournament. This was addressed by carefully aligning match dates, Elo ratings, and tournament start dates, and by using the Leave-One-Tournament-Out evaluation strategy to maintain strict temporal separation between training and testing data. Additional challenges included reconstructing tournament-level results and group-stage compositions from the downloaded datasets, as well as handling differences in tournament formats across competitions. These issues were addressed through conservative data filtering, date-based feature aggregation, and the creation of dedicated datasets that ensure consistent and comparable inputs for all tournaments.

3.2 Tools

The implementation relies primarily on Pandas and NumPy for data manipulation, Scikit-learn for model training and evaluation (logistic regression and random forest), and XGBoost for gradient boosting models. Matplotlib and Seaborn are used for data visualization, including performance metrics, feature importance, and confusion matrices.

4 Codebase & Reproducibility

4.1 How to run

The project provides both a requirements.txt file and a Conda environment configuration to manage dependencies. To run the full pipeline, clone the repository and install the required dependencies using one of the following methods:

1st Methode:

```
git clone https://github.com/DanyRDG/Final-Project.git
cd Final-Project
conda env create -f environment.yml
conda activate football-project
```

2nd Methode:

pip install -r requirements.txt

4.2 Reproducibility features

Random states are fixed at 42 for all machine learning models to ensure reproducible training and evaluation results.

5 Results

5.1 Global performance by tournament

Having identified multinomial logistic regression as the most reliable model overall, we now examine its performance in more detail across different tournament types. The results reveal noticeable variation depending on the competition (see Figure 1). Performance is strongest for tournaments such as the Gold Cup and Copa América, where both stage accuracy and macro-3 accuracy are relatively high and the mean absolute error remains low. This suggests that outcomes in these tournaments are more predictable, likely due to more stable team hierarchies and larger skill gaps between the teams of those continents.

In contrast, performance is weaker for tournaments such as the FIFA World Cup and the UEFA Euro, where stage accuracy decreases and MAE increases. These competitions typically feature a higher concentration of elite teams and greater competitive balance, making precise stage-level predictions more challenging.

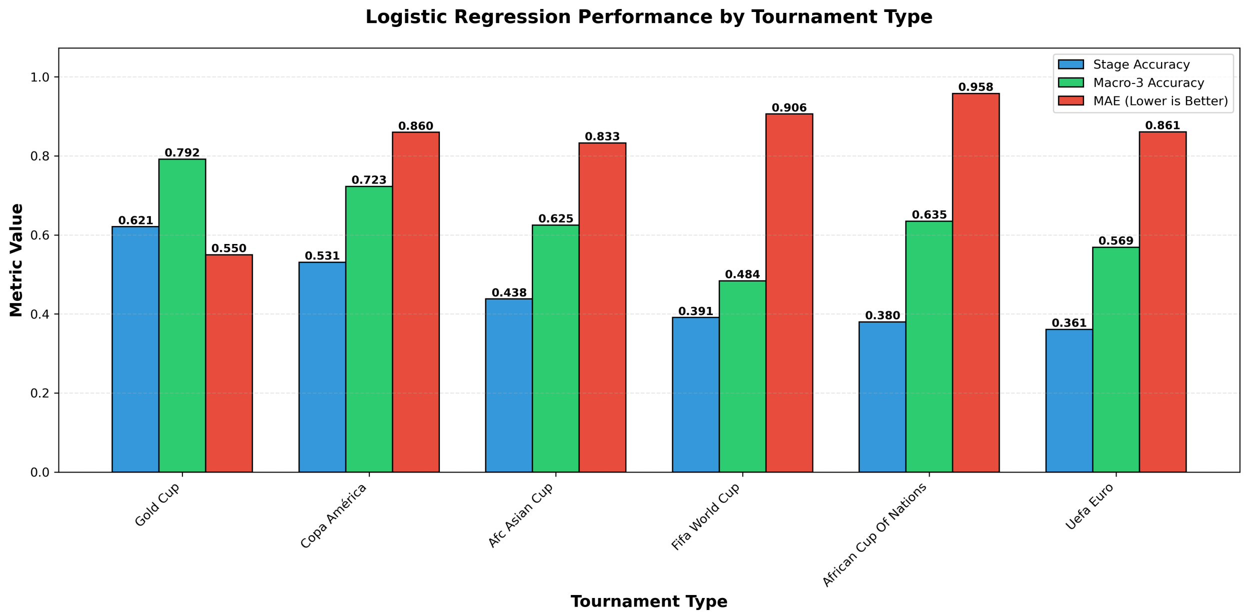
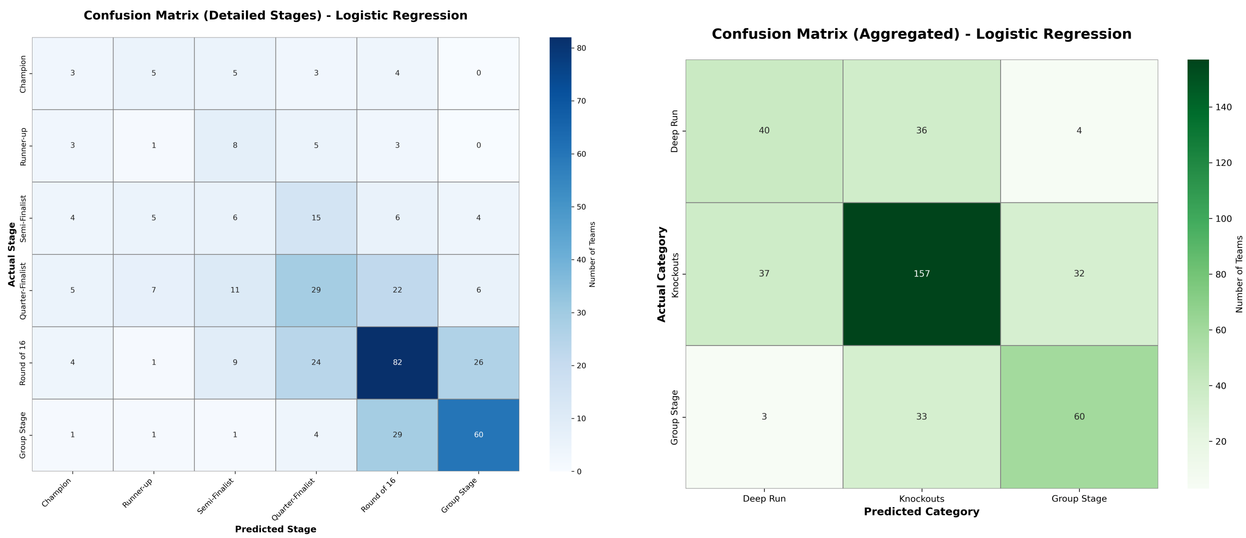


Figure 1: Logistic Regression performance by tournament

Overall, this analysis suggests that model performance is strongly influenced by tournament structure and the level of competitive balance among participating teams. In tournaments with a dense concentration of elite nations, predicting exact stages becomes particularly difficult, and performance metrics deteriorate accordingly. While the model still captures some broad patterns of tournament progression since it beats the random guessing baseline as we will talk in the Discussion module, its predictions should be interpreted as probabilistic tendencies rather than precise forecasts, highlighting both the potential and the limitations of data-driven approaches in international football.

5.2 Confusion matrix

The detailed-stage confusion matrix (Figure 2a) illustrates how the logistic regression model performs across the six tournament stages. Correct predictions are concentrated along the diagonal, with particularly strong performance for Group Stage and Round of 16 outcomes since their misclassifications tend to occur between adjacent outcomes, such as predicting quarter final exit instead of a round of 16 run, rather than between extreme stages like champion and round of 16 elimination. This pattern aligns with the previously reported MAE values being < 1 (see figure 1), indicating that when the model makes errors, it generally places teams within a close range of their true tournament outcome rather than making large misjudgments.



(a) Detailed stage confusion matrix

(b) Aggregated category confusion matrix

Figure 2: Confusion matrices for the Logistic Regression LOTO model

The aggregated confusion matrix (Figure 2b), which groups outcomes into Deep Run, Knockouts, and Group Stage, further highlights the robustness of the model at a broader level. Most teams are correctly assigned to their corresponding macro3 category, with errors primarily occurring between neighboring groups (for example, Deep Run versus Knockouts). These results suggest that while predicting exact finishing positions remains challenging, particularly among top-performing teams, the model is effective at capturing meaningful differences in overall tournament success. This supports the interpretation of the model as a reliable tool for probabilistic forecasting and comparative ranking rather than precise stage-level prediction.

5.3 Features importance

The feature importance analysis (see Figure 3) for the logistic regression model highlights which variables matter most when predicting tournament outcomes. Features with higher importance values have a stronger influence on the model's predictions, meaning they play a larger role in determining how far a team is expected to progress in a tournament. The graph shows that group-related metrics are the most influential drivers of the model's predictions, particularly a team's Elo rating relative to its group average. Measures of overall team strength, such as pre-tournament Elo, also play a significant role, while recent form and goal-based statistics contribute more modestly. This suggests that tournament outcomes are driven more by structural competitive context than by short-term performance fluctuations.

This result is intuitive, as tournament progression is heavily influenced by the relative strength of opponents faced in the group stage. Teams placed in weaker groups benefit from a clearer path to the knockouts stages, while strong teams in highly competitive groups face a higher risk of early elimination.

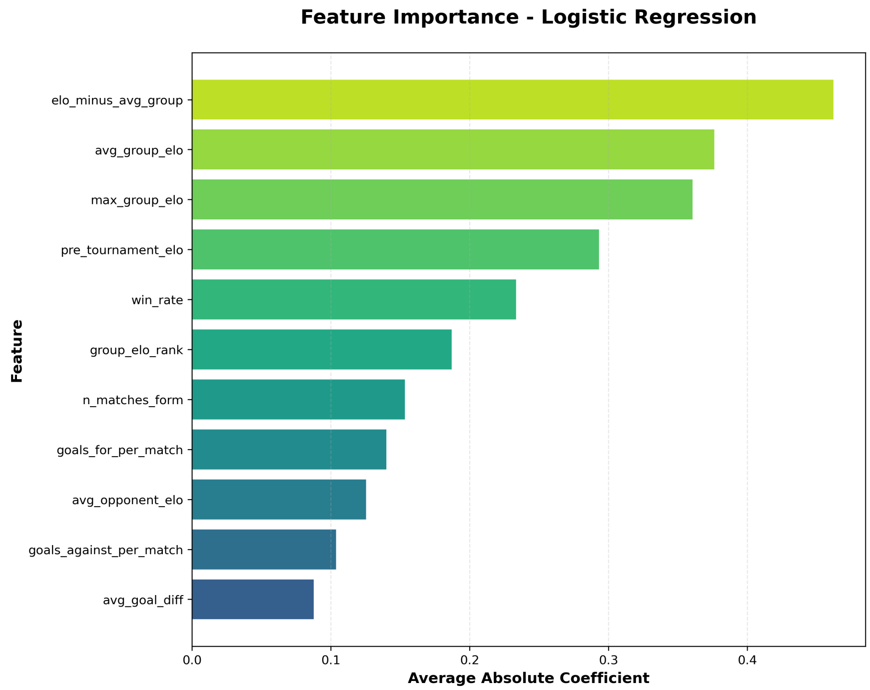


Figure 3: Feature importance for the logistic regression model

5.4 World cup 2026 predictions

After building and evaluating the modeling framework, the analysis now moves to a practical application. The best model (LogReg) is used to generate probabilistic predictions for the 2026 FIFA World Cup. The figure 4 ranks the top ten teams according to their tournament success score, which combines predicted probabilities across all tournament stages. Spain leads this ranking as favourites, followed by Brazil, Argentina, and France, suggesting that these teams are expected to perform consistently well throughout the tournament rather than simply having a high chance of winning the final.

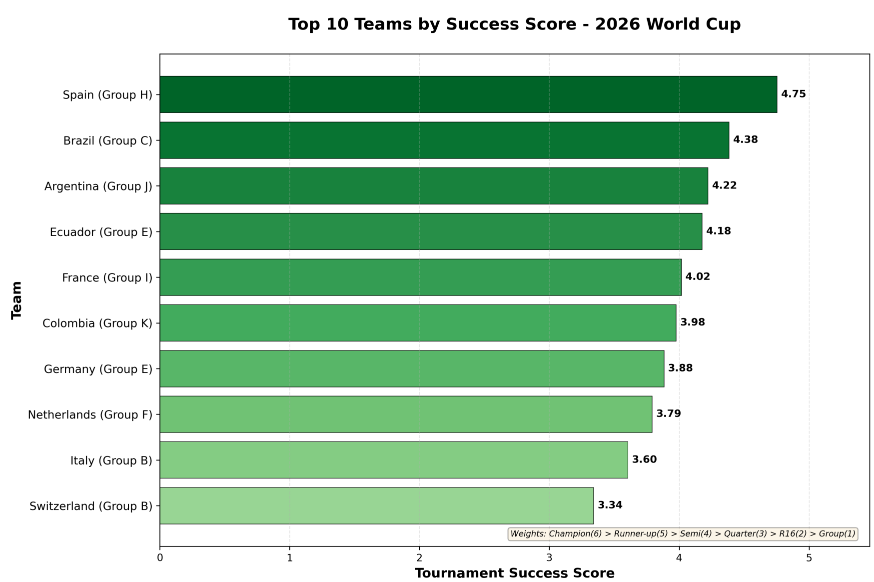


Figure 4: Top 10 teams by tournament success score for the 2026 FIFA World Cup

The figure 6 (see Appendix) displays the top ten teams by predicted champion probability for the 2026 FIFA World Cup. While the same set of traditionally strong teams appears near the top, the champion probabilities remain relatively moderate, with no team approaching certainty. This spread highlights the high level of uncertainty inherent in knockout tournaments and reinforces the idea that these predictions should be interpreted as likelihoods rather than definitive outcomes. Notably, both prediction graphs feature only teams from Europe and South America, indicating that the model successfully captures the historical dominance of these continents, as all past World Cup winners have come from Europe or South America.

6 Discussion

This section discusses the main findings of the project, focusing on model performance, alignment with expectations, and key limitations of predicting international football tournament outcomes using pre-tournament data.

6.1 What worked well

First, the results showed that pre-tournament features contain meaningful predictive information. Across tournaments, all models outperformed a random guessing baseline, indicating that the feature engineering approach captures relevant aspects of team strength and competitive context. For a World Cup–style format, the random guessing baseline achieves about 33.3% stage accuracy (and between 26% and 33.3% for some tournaments with other format), whereas the Logistic Regression reaches stage accuracies ranging from 36.1% to 62.1% (Figure 1). A similar improvement is observed for the macro-3 classification task, where the baseline accuracy is 40.6% compared to model accuracies between 48.4% and 79.2%. Secondly, the average prediction error remains relatively low across all tournaments, with the mean absolute error consistently below one stage. This indicates that when the model makes incorrect predictions, it typically places teams only one stage away from their actual outcome, such as predicting a semi-final exit instead of a quarter-final exit. Such errors are reasonable given the structure of knockout tournaments, where a single match can determine progression. A low average error therefore suggests that the model is effective at estimating a team's general tournament depth Thirdly, using probabilistic predictions instead of fixed stage labels proves to be an important advantage. By estimating probabilities for each possible tournament stage individually by team, the model captures uncertainty and allows teams to be ranked more realistically rather than forcing a single outcome. Finally, the Leave-One-Tournament-Out evaluation strategy works well in practice. It closely reflects a real prediction scenario by ensuring that each tournament is predicted using only past data, which helps prevent data leakage and provides a fair assessment of model performance across different competitions.

6.2 Limitations of the approach

Despite its contributions, the project has several important limitations. International football tournaments, that follow the cup style of tournaments, are characterized by a high degree of randomness, as outcomes are determined by a very small number of matches. Unlike league competitions, the knockout format amplifies the impact of a single match loss, making outcomes more volatile and less predictable from pre-tournament information alone. In addition, national teams play relatively few matches each year, which limits the amount of recent and consistent data available to accurately measure form and team strength. Changes in squad selection, injuries, or coaching decisions like the tactics can significantly affect performance but are not captured in the available data. Together, these factors show the uncertainty of tournament prediction and set clear boundaries on the reliability of this approach.

6.3 Comparison with Expectations

Observed performance aligns with expectations from football analytics literature, where moderate accuracy is typical when relying solely on pre-tournament information. The stronger generalization of logistic regression compared to more complex models suggests that simpler approaches are better suited to the available data.

Lower reliability in macro-level predictions for elite tournaments was anticipated, since strong teams tend to have similar levels of performance, making outcomes harder to distinguish.

7 Conclusion

This work explored whether pre-tournament data can be used to predict how national teams perform in major international football tournaments. By combining historical match results, Elo ratings, recent form indicators, and group-level contextual features, a structured and reproducible modeling framework is developed and evaluated across multiple competitions.

7.1 Summary of results

The goal of this project was to determine whether pre-tournament data could be used to predict how national teams perform in major international football tournaments. By combining different features, several predictive models were trained and evaluated using a Leave-One-Tournament-Out strategy. The results show that these features contain meaningful predictive signal, with the best model consistently outperforming a random baseline and achieving moderate accuracy across different competitions. However, exact stage-level prediction remains difficult, especially in highly competitive tournaments, where small differences between teams and knockout formats increase uncertainty.

In practice, the models tend to place teams within a reasonable range of outcomes rather than precisely identifying their final stage, which aligns with the inherently unpredictable nature of tournament football. This behavior is reflected in the relatively low mean absolute error, indicating that incorrect predictions are often close to the true outcome. While the models cannot fully eliminate uncertainty, they provide a structured and data-driven way to assess relative team strength before a tournament begins. Overall, the findings suggest that pre-tournament data can provide useful probabilistic insights, but should be interpreted as guidance rather than definitive forecasts.

7.2 Future Work

Future improvements could focus on enriching the feature set with additional contextual information such as squad composition, injuries, and coach tactics, which can strongly affect team performance but are not captured by match history alone. In addition, incorporating known tournament brackets after the group stage could improve predictions by accounting for potential opponent paths, as this information is available before the tournament begins and directly influences a team's chances of advancing.

8 References

1. FIFA.com. (2025). FIFA World Cup 2026 (Canada/Mexico/USA) [Official tournament page]. FIFA. https://www.fifa.com/en/tournaments/mens/worldcup/canadamexicousa2026

2. martj42. (n.d.). International Football Results from 1872 to 2017 [Dataset]. Kaggle. https://www.kaggle.com/datasets/martj42/international-football-results-from-1872-to-2017

3. martj42. (n.d.). International Football Results — results.csv [Dataset file]. GitHub. https://raw.githubusercontent.com/martj42/international_results/master/results.csv

4. martj42. (n.d.). International Football Results — shootouts.csv [Dataset file]. GitHub. https://raw.githubusercontent.com/martj42/international_results/master/shootouts.csv

9 Appendix

9.1 Random guessing baseline calculation

Take last world cup as an example, 32 teams divided in stage reached:

- 1 Champion
- 1 Runner-up
- 2 Semi-finalists
- 4 Quarter-finalists
- 8 Round of 16
- 16 Group stage

Notations
Let N denote the total number of teams.
Let there be exit-round categories indexed by $i \in \{\text{Champion}, \dots, \text{Group Stage}\}$
Let

$$k_i = \text{number of teams that end in category } i$$

Since each team is assigned to exactly one category, we have

$$\sum_{i=\text{Champion}}^{\text{Group Stage}} k_i = N$$

Variable
For each team t, define the indicator variable

$$X_t = \begin{cases} 1, & \text{if team } t \text{ is predicted correctly,} \\ 0, & \text{otherwise} \end{cases}$$

Total number of correctly predicted teams
The total number of correct predictions is given by

$$X = \sum_{t=1}^{N} X_t$$

Expected value for a single team
Suppose team t truly belongs to category i, which contains k_i teams in total.
Assuming that all N tournament slots are equally likely, the expected value of X_t is

$$\mathbb{E}[X_t] = \mathbb{P}(X_t = 1) = \frac{k_i}{N}$$

Expected number of correctly predicted teams from category i
There are k_i teams in category i, each with expected value $\mathbb{E}[X_t] = \frac{k_i}{N}$.
Therefore, the expected number of correctly predicted teams from category i is

$$\mathbb{E}[\text{correct teams from category } i] = k_i \cdot \frac{k_i}{N} = \frac{k_i^2}{N}$$

Total expected number of correctly predicted teams
Summing over all tournament categories, the total expected number of correct predictions is

$$\mathbb{E}[X] = \sum_{i=\text{Champion}}^{\text{Group Stage}} \frac{k_i^2}{N}$$

For the six-stage tournament structure considered in this study, this yields

$$\mathbb{E}[X] = 33.3\%$$

Aggregation into three categories
The same calculation can be repeated after aggregating tournament outcomes into three categories: Deep Run, Knockouts, and Group Stage. In this case,

$$\mathbb{E}[X] = \sum_{i=\text{Deep Run}}^{\text{Group Stage}} \frac{k_i^2}{N}$$

Under this coarser categorization, the expected proportion of correct predictions increases to

$$\mathbb{E}[X] = 40.6\%$$

9.2 Additional graphs

predicted_stage_label
Champion
Runner-up
Semi-Finalist
Semi-Finalist
Quarter-Finalist
Quarter-Finalist
Quarter-Finalist
Quarter-Finalist
Round of 16
Round of 16
Round of 16
Round of 16
Round of 16
Round of 16
Round of 16
Round of 16
Group Stage
Group Stage
Group Stage
Group Stage
Group Stage

(a) Six-stage outcome model

pred_macro_label
Deep Run
Deep Run
Deep Run
Deep Run
Knockouts
Knockouts
Knockouts
Knockouts
Knockouts
Knockouts
Knockouts
Knockouts
Knockouts
Knockouts
Knockouts
Knockouts
Group Stage
Group Stage
Group Stage
Group Stage
Group Stage

(b) Macro-3 outcome model

Figure 5: Outcome modeling schemes used in this study

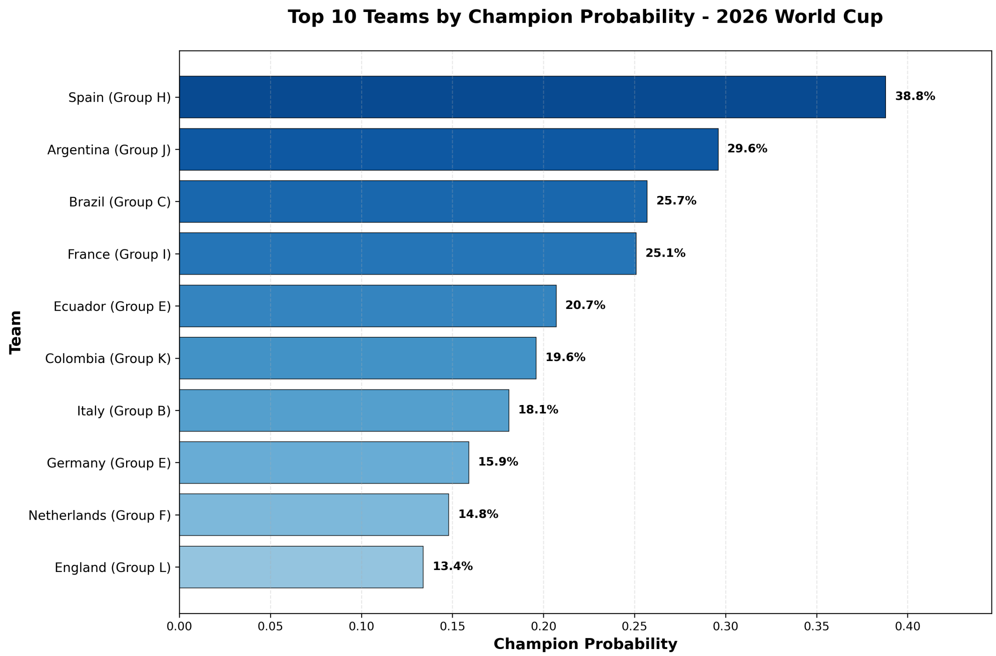


Figure 6: Top 10 champion probability for the World Cup 2026

9.3 AI Disclosure

AI tools were used during the development of this project to support specific technical and documentation-related tasks. ChatGPT and Claude were used to assist with code debugging, writing and refining scripts for downloading external datasets, and helping to rework the environment.yml file after initial execution issues were identified when other users attempted to run the project. These tools were also used to support parts of the project documentation, mainly to improve clarity and structure.

All key analytical decisions in this project were made by the author. AI tools were used only as support and were carefully reviewed and adapted before use. The final implementation and interpretations reflect the author's own work and judgment.

9.4 Project structure

```
Final-Project/
├── main.py                              # Main entry point (runs full pipeline)
├── environment.yml                      # Conda environment specification
├── requirements.txt                     # pip dependencies
├── README.md                            # Project documentation
├── PROPOSAL.md                          # Initial project proposal
│
├── src/                                 # Source code
│   ├── __init__.py                      # Package initializer
│   ├── data_loading_cleaning.py         # Raw data loading and cleaning
│   ├── elo_calculator.py                # Elo rating computation
│   ├── features.py                      # Feature engineering
│   ├── group_stage_composition_builder.py # Group-stage reconstruction
│   ├── models.py                        # Model training and evaluation
│   ├── tournament_builder.py            # Tournament structure creation
│   ├── tournament_results_builder.py    # Tournament results processing
│   ├── Visualization.py                 # Data visualization utilities
│   └── World_Cup2026.py                 # 2026 World Cup predictions
│
├── data/                                # Data directory
│   ├── raw/                             # Raw, unprocessed data
│   │   ├── results.csv
│   │   └── shootouts.csv
│   │
│   └── processed/                       # Processed datasets (generated)
│
├── results/                             # Model outputs
│   ├── models/                          # Trained models and scalers
│   ├── predictions/                     # Prediction outputs
│   └── visualizations/                  # Figures and plots
```

Figure 7: Project structure